

Importing Libraries

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
data=pd.read_csv("/Heart_Attack_Risk_Levels_Dataset.csv")
```

Exploring Dataset

data

	Age	Gender	Heart rate	Systolic blood pressure	Diastolic blood pressure	Blood sugar	CK-MB	Troponin	Result	Risk_Level	Recommendation
0	63	1	66	160	83	160.0	1.80	0.012	negative	Moderate	Monitor closely and consult doctor
1	20	1	94	98	46	296.0	6.75	1.060	positive	High	Immediate medical attention
2	56	1	64	160	77	270.0	1.99	0.003	negative	Moderate	Monitor closely and consult doctor
3	66	1	70	120	55	270.0	13.87	0.122	positive	High	Immediate medical attention
4	54	1	64	112	65	300.0	1.08	0.003	negative	Moderate	Monitor closely and consult doctor
...	...	...	...	...	...	...	...	...	...	...	...
1314	44	1	94	122	67	204.0	1.63	0.006	negative	Moderate	Monitor closely and consult doctor
1315	66	1	84	125	55	149.0	1.33	0.172	positive	High	Immediate medical attention
1316	45	1	85	168	104	96.0	1.24	4.250	positive	High	Immediate medical attention
1317	54	1	58	117	68	443.0	5.80	0.359	positive	High	Immediate medical attention
1318	51	1	94	157	79	134.0	50.89	1.770	positive	High	Immediate medical attention

1319 rows × 11 columns

data.head()

	Age	Gender	Heart rate	Systolic blood pressure	Diastolic blood pressure	Blood sugar	CK-MB	Troponin	Result	Risk_Level	Recommendation
0	63	1	66	160	83	160.0	1.80	0.012	negative	Moderate	Monitor closely and consult doctor
1	20	1	94	98	46	296.0	6.75	1.060	positive	High	Immediate medical attention
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Next steps: [Generate code with data](#) [View recommended plots](#) [New interactive sheet](#)

data.tail()

	Age	Gender	Heart rate	Systolic blood pressure	Diastolic blood pressure	Blood sugar	CK-MB	Troponin	Result	Risk_Level	Recommendation
1314	44	1	94	122	67	204.0	1.63	0.006	negative	Moderate	Monitor closely and consult doctor
1315	66	1	84	125	55	149.0	1.33	0.172	positive	High	Immediate medical attention
1316	45	1	85	168	104	96.0	1.24	4.250	positive	High	Immediate medical attention
1317	54	1	58	117	68	443.0	5.80	0.359	positive	High	Immediate medical

data.info()

<class 'pandas.core.frame.DataFrame'> RangeIndex: 1319 entries, 0 to 1318
--

Data columns (total 11 columns):

#	Column	Non-Null	Count	Dtype
0	Age	1319	non-null	int64
1	Gender	1319	non-null	int64
2	Heart rate	1319	non-null	int64
3	Systolic blood pressure	1319	non-null	int64
4	Diastolic blood pressure	1319	non-null	int64
5	Blood sugar	1319	non-null	float64
6	CK-MB	1319	non-null	float64
7	Troponin	1319	non-null	float64
8	Result	1319	non-null	object
9	Risk_Level	1319	non-null	object
10	Recommendation	1319	non-null	object

dtypes: float64(3), int64(5), object(3)
memory usage: 113.5+ KB

data.shape

(1319, 11)

data.nunique()

	0
Age	75
Gender	2
Heart rate	79
Systolic blood pressure	116
Diastolic blood pressure	73
Blood sugar	244
CK-MB	700
Troponin	352
Result	2
Risk_Level	3
Recommendation	3

dtype: int64

data.columns

Index(['Age', 'Gender', 'Heart rate', 'Systolic blood pressure', 'Diastolic blood pressure', 'Blood sugar', 'CK-MB', 'Troponin', 'Result', 'Risk_Level', 'Recommendation'], dtype='object')

data.dtypes

	0
Age	int64
Gender	int64
Heart rate	int64
Systolic blood pressure	int64
Diastolic blood pressure	int64
Blood sugar	float64
CK-MB	float64
Troponin	float64
Result	object
Risk_Level	object
Recommendation	object

dtype: object

data.describe()

	Age	Gender	Heart rate	Systolic blood pressure	Diastolic blood pressure	Blood sugar	CK-MB	Troponin	
count	1319.000000	1319.000000	1319.000000	1319.000000	1319.000000	1319.000000	1319.000000	1319.000000	
mean	56.193328	0.659591	78.336619	127.170584	72.269143	146.634344	15.274306	0.360942	
std	13.638173	0.474027	51.630270	26.122720	14.033924	74.923045	46.327083	1.154568	
min	14.000000	0.000000	20.000000	42.000000	38.000000	35.000000	0.321000	0.001000	
25%	47.000000	0.000000	64.000000	110.000000	62.000000	98.000000	1.655000	0.006000	
50%	58.000000	1.000000	74.000000	124.000000	72.000000	116.000000	2.850000	0.014000	
75%	65.000000	1.000000	85.000000	143.000000	81.000000	169.500000	5.805000	0.085500	
max	103.000000	1.000000	1111.000000	223.000000	154.000000	541.000000	300.000000	10.300000	

Start coding or [generate](#) with AI.

Checking Missing and Duplicate Values

	Age	Gender	Heart rate	Systolic blood pressure	Diastolic blood pressure	Blood sugar	CK-MB	Troponin	Result	Risk_Level	Recommendation
0	63	1	66	160	83	160.0	1.80	0.012	negative	Moderate	Monitor closely and consult doctor
1	20	1	94	98	46	296.0	6.75	1.060	positive	High	Immediate medical attention
2	56	1	64	160	77	270.0	1.99	0.003	negative	Moderate	Monitor closely and consult doctor
3	66	1	70	120	55	270.0	13.87	0.122	positive	High	Immediate medical attention
4	54	1	64	112	65	300.0	1.08	0.003	negative	Moderate	Monitor closely and consult doctor

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	0
Age	0
Gender	0
Heart rate	0
Systolic blood pressure	0
Diastolic blood pressure	0
Blood sugar	0
CK-MB	0
Troponin	0
Result	0
Risk_Level	0
Recommendation	0

dtype: int64

data.notnull().sum()



	0
Age	1319
Gender	1319
Heart rate	1319
Systolic blood pressure	1319
Diastolic blood pressure	1319
Blood sugar	1319
CK-MB	1319
Troponin	1319
Result	1319
Risk_Level	1319
Recommendation	1319

dtype: int64

```
data[data.duplicated()]
```

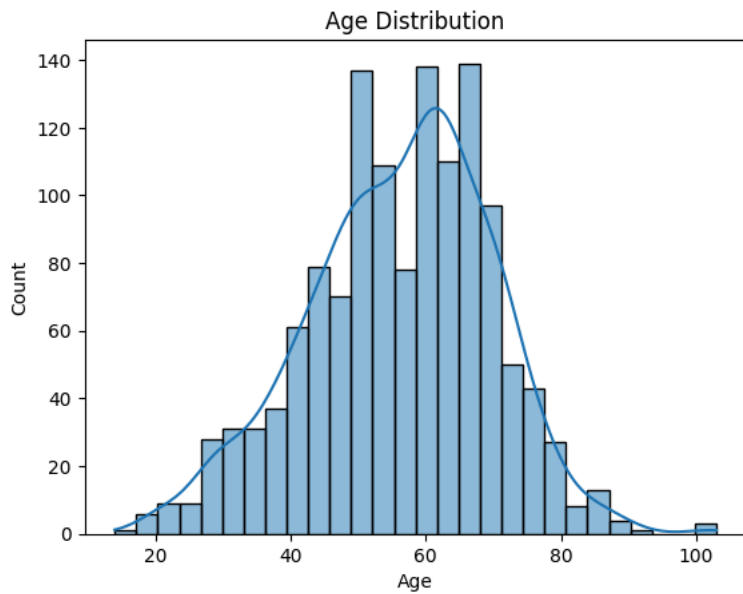


Age	Gender	Heart rate	Systolic blood pressure	Diastolic blood pressure	Blood sugar	CK-MB	Troponin	Result	Risk_Level	Recommendation
-----	--------	------------	-------------------------	--------------------------	-------------	-------	----------	--------	------------	----------------



▼ Age Distribution

```
sns.histplot(data=data, x='Age', kde=True)
plt.title('Age Distribution')
plt.show()
```



▼ Age by Risk Level

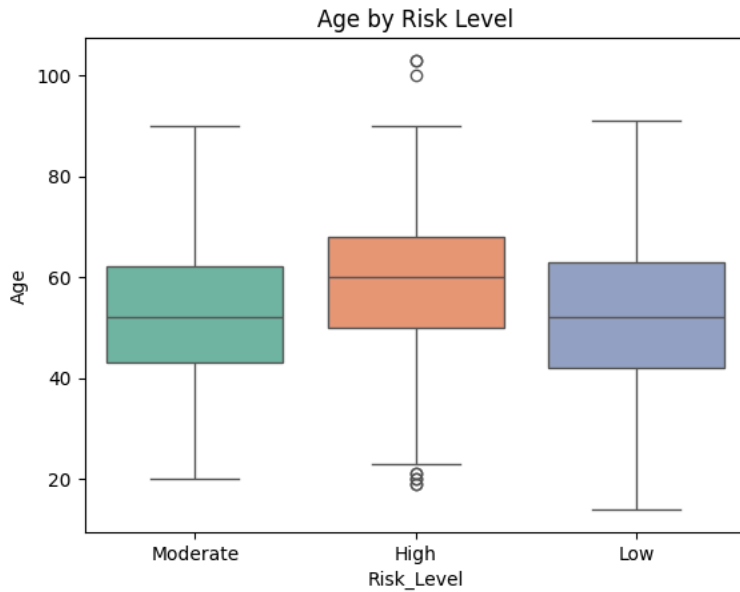
```
sns.boxplot(data=data, x='Risk_Level', y='Age', palette='Set2')
plt.title('Age by Risk Level')
plt.show()
```



<ipython-input-57-3240f977c3d4>:1: FutureWarning:

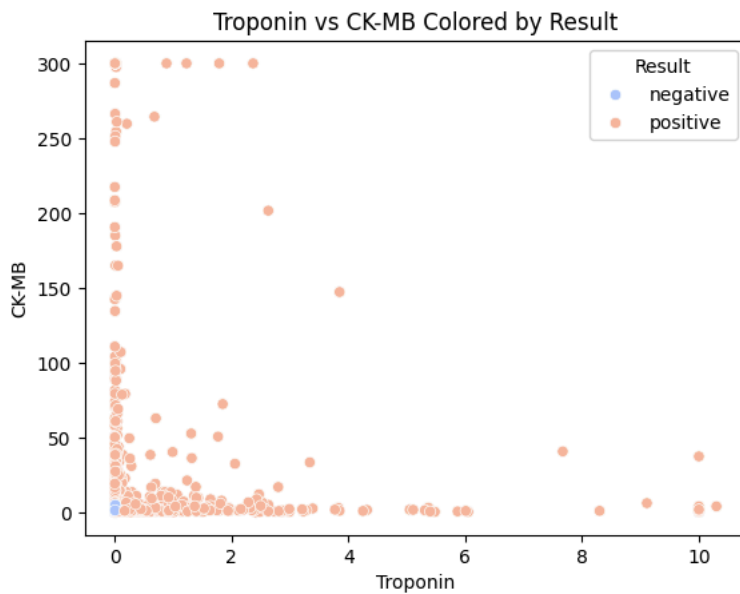
Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False`.

```
sns.boxplot(data=data, x='Risk_Level', y='Age', palette='Set2')
```



✓ Troponin vs CK-MB Colored by Result

```
sns.scatterplot(data=data, x='Troponin', y='CK-MB', hue='Result', palette='coolwarm')
plt.title('Troponin vs CK-MB Colored by Result')
plt.show()
```

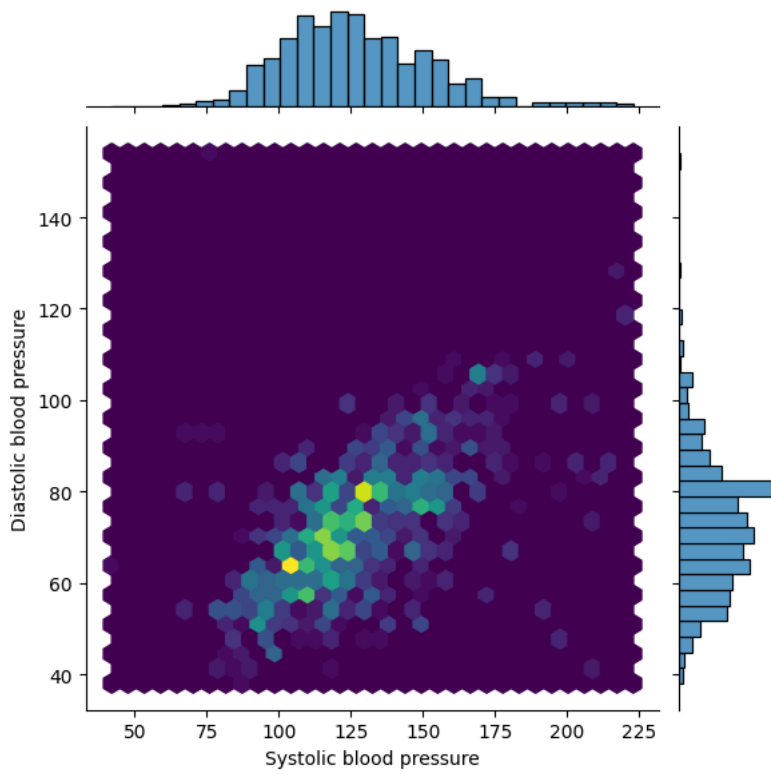


✓ Joint Distribution of Blood Pressure

```
sns.jointplot(data=data, x='Systolic blood pressure', y='Diastolic blood pressure', kind='hex', cmap='viridis')
plt.suptitle('Joint Distribution of Blood Pressure', y=1.02)
plt.show()
```



Joint Distribution of Blood Pressure

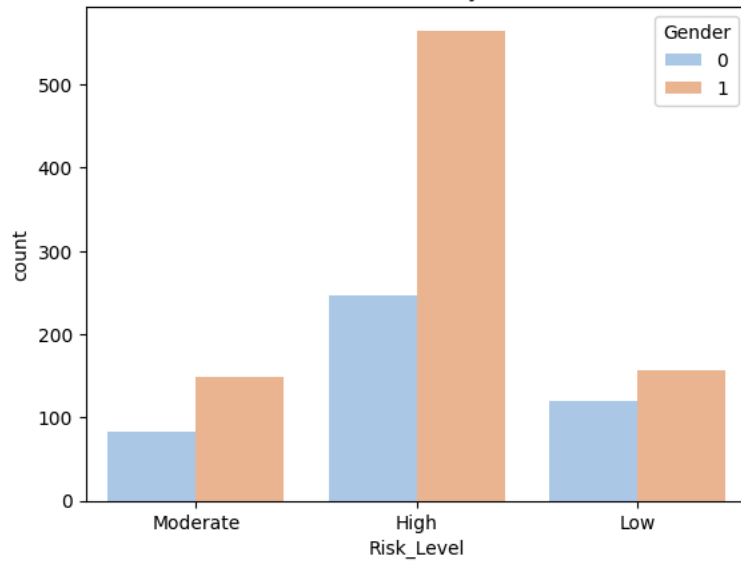


✓ Risk Level Count by Gender

```
sns.countplot(data=data, x='Risk_Level', hue='Gender', palette='pastel')  
plt.title('Risk Level Count by Gender')  
plt.show()
```



Risk Level Count by Gender



✓ Correlation Heatmap

```
plt.figure(figsize=(10, 8))  
sns.heatmap(data corr(numeric_only=True), annot=True, cmap='coolwarm', fmt=".2f")  
plt.title('Correlation Heatmap')  
plt.show()
```



Correlation Heatmap

